



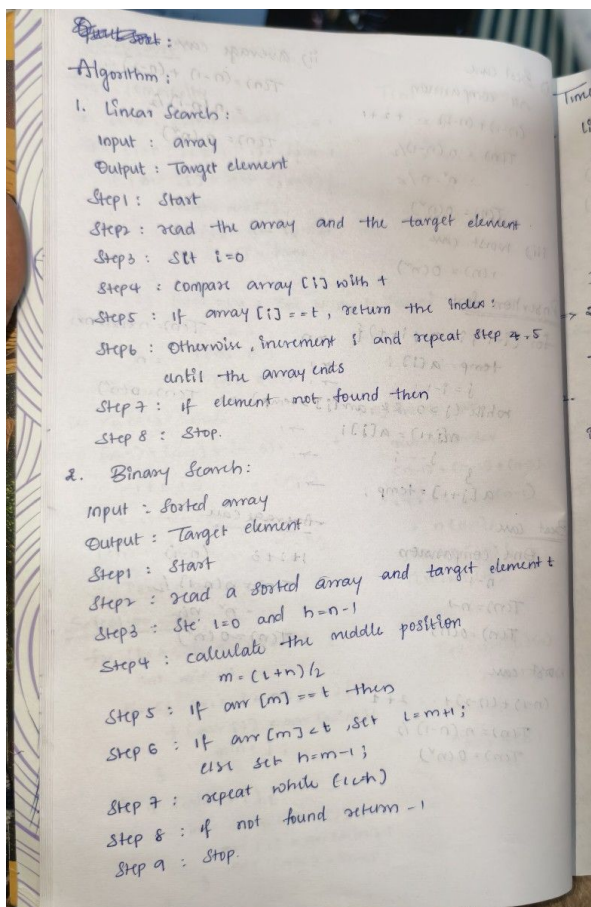
Practical: 2

Implementation and Time analysis of linear and binary search algorithm

05/08/2026

Aim: To Implement Linear search and binary search and Time analysis their algorithms.

Algorithm:



Program:

```
#include <iostream>
using namespace std;
```

```
// Linear Search
```

```
int linear(int array[], int s, int t) {
    for (int i = 0; i < s; i++) {
        if (array[i] == t) {
            return i;
        }
    }
}
```



```
    }
    return -1;
}

// Binary Search
int binary(int array[], int s, int t) {
    int l = 0;
    int h = s - 1;

    while (l <= h) {
        int m = (l + h) / 2;

        if (array[m] == t) {
            return m;
        }
        else if (array[m] < t) {
            l = m + 1;
        }
        else {
            h = m - 1;
        }
    }

    return -1;
}

// Display Array
void display(int array[], int n) {
    cout << "Array is:\n";

    for (int i = 0; i < n; i++) {
        cout << array[i] << " ";
    }

    cout << endl;
}

int main() {

    int array[] = {2, 6, 7, 9, 11};
    int n = sizeof(array) / sizeof(array[0]);

    int t = 6;

    display(array, n);

    int f = linear(array, n, t);
    int g = binary(array, n, t);
}
```



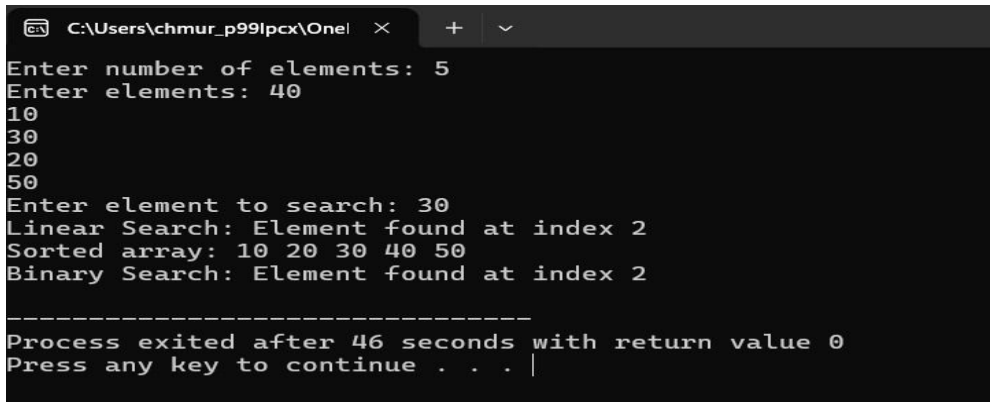
```
cout << "\nTarget element: " << t << endl;

if (f != -1) {
    cout << "Linear Search: Element found at index " << f << endl;
}
else {
    cout << "Linear Search: Element not found" << endl;
}

if (g != -1) {
    cout << "Binary Search: Element found at index " << g << endl;
}
else {
    cout << "Binary Search: Element not found" << endl;
}

return 0;
}
```

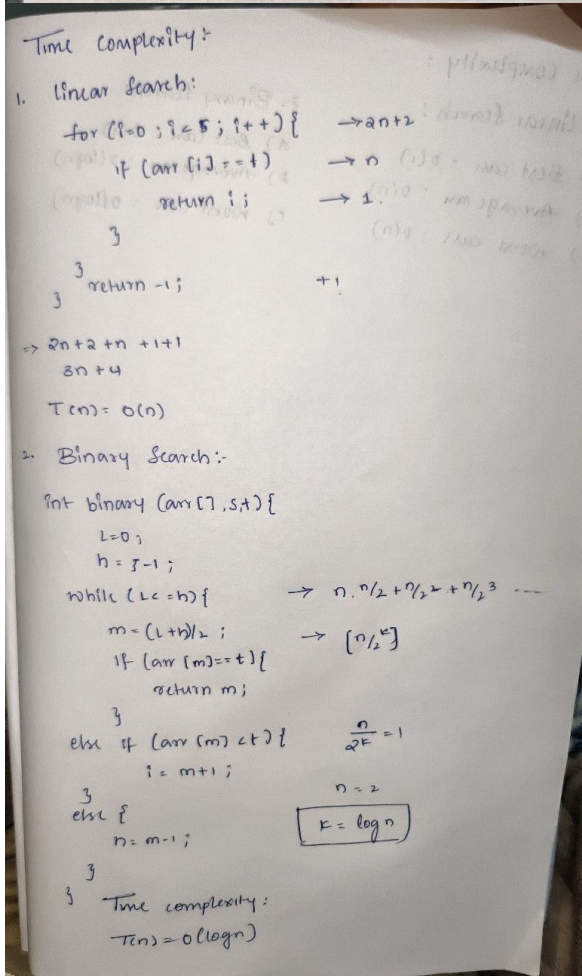
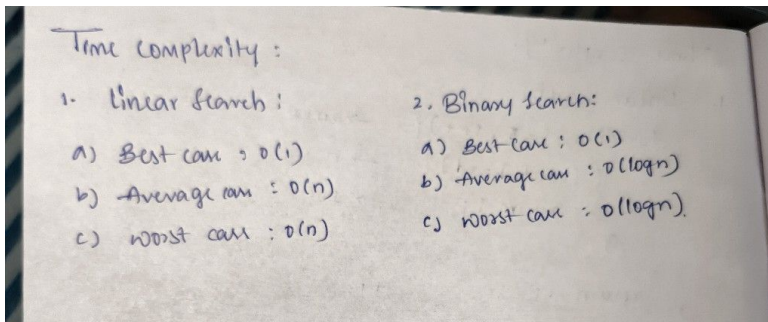
Output:



```
C:\Users\chmur_p99lpcx\One
Enter number of elements: 5
Enter elements: 40
10
30
20
50
Enter element to search: 30
Linear Search: Element found at index 2
Sorted array: 10 20 30 40 50
Binary Search: Element found at index 2

-----
Process exited after 46 seconds with return value 0
Press any key to continue . . . |
```

Time Complexity:



Conclusion:

Hence

From this practical of linear search, Binary search the algorithm and programs have executed successfully. And Time complexity has categorised into three ways as Best, average and worst.